

WSP - Web System Programming

Laboratory 06 – JS 2

JavaScript basics - part 2

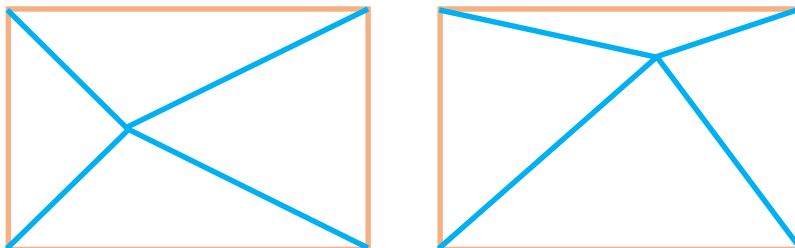
Do the following tasks.

1. Create a website where, **using JavaScript** and methods for creating DOM elements, a multiplication table will be created for n random elements in the range of 1-99. The number n (number of rows and columns) with the results of multiplication should be given by the user. The value of n is to be of the range ($5 \leq n \leq 20$). If the user provides **anything else**, some **default value** from this range is to be assumed, but the page is **to indicate** that the given data was incorrect, so $n=X$ was assumed. The values drawn are to be included in the column and row headings. Even results (multiplications) in array cells should be with the class "even" and odd ones with the class "odd". In the CSS file, style the array looks aesthetically pleasing and visually distinguishes the parity of the results. **E.g. $n=13$:**

	55	91	51	93	94	40	69	50	86	6	47	19	65
55	3025	5005	2805	5115	5170	2200	3795	2750	4730	330	2585	1045	3575
91	5005	8281	4641	8463	8554	3640	6279	4550	7826	546	4277	1729	5915
51	2805	4641	2601	4743	4794	2040	3519	2550	4386	306	2397	969	3315
93	5115	8463	4743	8649	8742	3720	6417	4650	7998	558	4371	1767	6045
94	5170	8554	4794	8742	8836	3760	6486	4700	8084	564	4418	1786	6110
40	2200	3640	2040	3720	3760	1600	2760	2000	3440	240	1880	760	2600
69	3795	6279	3519	6417	6486	2760	4761	3450	5934	414	3243	1311	4485
50	2750	4550	2550	4650	4700	2000	3450	2500	4300	300	2350	950	3250
86	4730	7826	4386	7998	8084	3440	5934	4300	7396	516	4042	1634	5590
6	330	546	306	558	564	240	414	300	516	36	282	114	390
47	2585	4277	2397	4371	4418	1880	3243	2350	4042	282	2209	893	3055
19	1045	1729	969	1767	1786	760	1311	950	1634	114	893	361	1235
65	3575	5915	3315	6045	6110	2600	4485	3250	5590	390	3055	1235	4225

All HTML elements **create using** `document.createElement()` method, to set HTML attributes **use DOM elements properties**. **Don't use** `document.write/writeline` methods.

2. Read about the `<canvas>` element. Create a canvas where mouse movement will draw one line from each corner to the mouse position. After moving the mouse, these lines will lead to the next place:



If the mouse cursor leaves the element, the lines are supposed to disappear. The JavaScript code is supposed to work correctly even if we change the dimensions of the `<canvas>` object

(e.g. in the styles of this object). The solution should allow you to insert the `<canvas>` element anywhere in the correct document and **insert several** (it is good to add a class to such a HTML element, e.a. `"drawingX"` and add listeners for all such elements) correctly working such elements. Add some text on page, so the page will be longer than one screen.

Deadline 1: meeting 8 (max 100 points)

Deadline 2: meeting 9 (max 80 points)

Deadline 3: meeting 10 (max 50 points)